量子與統計

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1. 電子自旋的依據，Stern-Gerlach 實驗

電子自旋的描述：$\vec{S}^2=(s+1)s\hbar^2$，s是自旋量子數

$S_z=\pm\frac{\hbar}{2}$

$\hat{\vec{S}}\times\hat{\vec{S}}=i\hbar\hat{\vec{S}}$，$[\hat{S}_\alpha,\hat{S}_\beta]=i\hbar\varepsilon_{\alpha\beta\gamma}\hat{S}_\gamma$，$[\hat{\vec{S}}^2,\hat{S}_z]=0$

共同本征態 $|Sm\rangle$，$\hat{\vec{S}}^2|Sm\rangle=s(s+1)\hbar^2$，$s=0,\frac{1}{2},1,\frac{3}{2},\cdots$

$\hat{S}_z|Sm\rangle=m\hbar$　　$m=-s,-s+1,\cdots,s-1,s$

$\hat{S}_x,\hat{S}_y,\hat{S}_z$ 的本征值均為 $\pm\frac{\hbar}{2}$，$\hat{S}_x^2=\hat{S}_y^2=\hat{S}_z^2=\frac{\hbar^2}{4}$

Pauli 算符 $\hat{S}_\alpha=\frac{\hbar}{2}\hat{\sigma}_\alpha$，$[\hat{\sigma}_\alpha,\hat{\sigma}_\beta]=2i\varepsilon_{\alpha\beta\gamma}\hat{\sigma}_\gamma$

$\hat{\sigma}_x\hat{\sigma}_y=-\hat{\sigma}_y\hat{\sigma}_x=i\hat{\sigma}_z$

$\hat{\sigma}_x=\begin{pmatrix}0&1\\1&0\end{pmatrix}$，$\hat{\sigma}_y=\begin{pmatrix}0&i\\-i&0\end{pmatrix}$　$\sigma_z=\begin{pmatrix}1&0\\0&1\end{pmatrix}$

電子自旋態：$\psi(\vec{r},t)=\begin{pmatrix}\psi_1(\vec{r},t)\\\psi_2(\vec{r},t)\end{pmatrix}\begin{matrix}\leftarrow S_z=\frac{\hbar}{2}\\\leftarrow S_z=-\frac{\hbar}{2}\end{matrix}$

若自旋與軌道相互作用小，$\psi(\vec{r},t)=\Psi_0(\vec{r},t)\chi(S_z)$

自旋磁矩：$|\mu_z|=\mu_B=\frac{e\hbar}{2m_e}$

$\hat{\vec{\mu}}_s=-2\cdot\frac{\mu_B}{\hbar}\vec{S}=-\frac{e}{m_e}\hat{\vec{S}}$

軌道磁矩 $\hat{\vec{\mu}}_L=-\frac{\mu_B}{\hbar}\hat{\vec{L}}$，$W=-(\hat{\vec{\mu}}_L+\hat{\vec{\mu}}_s)\cdot\vec{B}$

角動量合成：$[\hat{\vec{J}}_1,\hat{\vec{J}}_2]=0$，則 $\hat{\vec{J}}_1+\hat{\vec{J}}_2=\hat{\vec{J}}$ 也是角動量

$$\left\{\begin{array}{l}\hat{\vec{J}}^+=\hat{\vec{J}}\\ \hat{\vec{J}}\times\hat{\vec{J}}=i\hbar\hat{\vec{J}}\end{array}\right.$$

$[\hat{J}_\alpha,\hat{J}_\beta]=i\hbar\varepsilon_{\alpha\beta\gamma}\hat{J}_\gamma$　$\alpha,\beta,\gamma\in\{x,y,z\}$

$[\hat{\vec{J}}^2,\hat{\vec{J}}_1^2]=[\hat{\vec{J}}^2,\hat{\vec{J}}_2^2]=0$

耦合表象 $\{\hat{\vec{J}}_1^2,\hat{\vec{J}}_2^2,\hat{\vec{J}}^2,\hat{J}_z\}$　非耦合表象 $\{\hat{\vec{J}}_1^2,\hat{J}_{1z},\hat{\vec{J}}_2^2,\hat{J}_{2z}\}$

$|j_1j_2jm\rangle$　　$|j_1m_1\rangle|j_2m_2\rangle$

$(2j_1+1)(2j_2+1)$

全同粒子：全部內稟性質完全相同

Bose子，Fermi子

對稱　反對稱

雙電子的自旋函數：$\chi^{(1)}=|\uparrow\uparrow\rangle$，$\chi^{(2)}=|\downarrow\downarrow\rangle$，

$\chi^{(3)}=|\uparrow\downarrow\rangle$，$\chi^{(4)}=|\downarrow\uparrow\rangle$

$\chi_S=\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle+|\downarrow\uparrow\rangle)$，$\chi_A=\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle-|\downarrow\uparrow\rangle)$

量子與統計 [第三次作業]

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2. (7.1) 證：$\hat{\sigma}_x\hat{\sigma}_y\hat{\sigma}_z = \frac{1}{2}(2i\hat{\sigma}_z)\hat{\sigma}_z = i$

3. (7.2)

$$\overline{S_x} = (\chi_{\frac{1}{2}}(s_z), \hat{S}_x\chi_{\frac{1}{2}}(s_z)) = (\chi_{\frac{1}{2}}, \frac{\hbar}{4i}[\hat{\sigma}_y,\hat{\sigma}_z]\chi_{\frac{1}{2}})$$

$$= (\chi_{\frac{1}{2}}, \frac{1}{2i}[\hat{\sigma}_y,\hat{S}_z]\chi_{\frac{1}{2}}) = \frac{1}{2i}(\chi_{\frac{1}{2}}, \hat{\sigma}_y(\frac{\hbar}{2}\chi_{\frac{1}{2}}) - \hat{S}_z\hat{\sigma}_y\chi_{\frac{1}{2}})$$

$$= \frac{\hbar}{4i}\overline{\sigma_y} - \frac{1}{2i}(\hat{S}_z\chi_{\frac{1}{2}}, \hat{\sigma}_y\chi_{\frac{1}{2}}) = \frac{\hbar}{4i}\overline{\sigma_y} - \frac{\hbar}{4i}\overline{\sigma_y} = 0$$

同理 $\overline{S_y} = 0$

$$\hat{S}_x^2 = \hat{S}_y^2 = \frac{\hbar^2}{4} \Rightarrow \overline{S_x^2} = \overline{S_y^2} = \frac{\hbar^2}{4}$$

$$\overline{(\Delta S_x)^2} = \frac{\hbar^2}{4} - 0 = \frac{\hbar^2}{4} = \overline{\Delta S_y^2} \Rightarrow \overline{\Delta S_x^2}\;\overline{\Delta S_y^2} = \frac{\hbar^4}{16}$$

4. (7.3)

(1) $\hat{S}_x$：$\begin{vmatrix} 0-\sigma_x & 1 \\ 1 & 0-\sigma_x \end{vmatrix} = \sigma_x^2 - 1 = 0$

$\sigma_x = \pm 1$，因此 $\hat{S}_x$ 的特徵值是 $\pm\frac{\hbar}{2}$，本征函数是 $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \pm\frac{1}{\sqrt{2}} \end{pmatrix}$

(2) $\hat{S}_y$：$\begin{vmatrix} 0-\sigma_y & -i \\ i & 0-\sigma_y \end{vmatrix} = \sigma_y^2 - 1 = 0$

$\sigma_y = \pm 1$，因此 $\hat{S}_y$ 的本徵值是 $\pm\frac{\hbar}{2}$，本征函数是 $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$ 和 $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$，即 $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm i \end{pmatrix}$

5. (7.5)

(1) 記 $\Psi_{nlm}(r,\theta,\varphi) = R_{nl}(r)Y_{lm}(\theta,\varphi)$

$$\psi = \begin{pmatrix} \frac{1}{2}\Psi_{211} \\ -\frac{\sqrt{3}}{2}\Psi_{210} \end{pmatrix},\quad \hat{L}_z\psi = \begin{pmatrix} \frac{1}{2}\hbar\Psi_{210} \\ -\frac{\sqrt{3}}{2}\cdot 0\cdot\hbar\Psi_{210} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}\hbar\Psi_{211} \\ 0 \end{pmatrix}$$

$$\overline{L_z} = (\psi, \hat{L}_z\psi) = \int \psi^+\hat{L}_z\psi\, d\tau = \frac{\hbar}{4}$$

$$\hat{S}_z\psi = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \frac{1}{2}\Psi_{211} \\ -\frac{\sqrt{3}}{2}\Psi_{210} \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} \frac{1}{2}\Psi_{211} \\ \frac{\sqrt{3}}{2}\Psi_{210} \end{pmatrix}$$

$$\overline{S_z} = \frac{\hbar}{2}\left[(\tfrac{1}{2})^2 - (\tfrac{\sqrt{3}}{2})^2\right] = -\frac{\hbar}{4}$$

(2) $\overline{M_z} = -\frac{e}{2m}\overline{L_z} - \frac{e}{m}\overline{S_z} = -\frac{e}{2m}(2\overline{L_z} + \overline{S_z}) = -\frac{e}{2m}(\frac{\hbar}{2} - \frac{\hbar}{4}) = -\frac{e\hbar}{8m}$

6. (7.7) 設兩個單粒子態波函數是 Ψ_1, Ψ_2，三個全同玻色子的坐標及自旋變量為 q_1, q_2, q_3

三個玻色子位於同一態上：$\Psi_1(q_1)\Psi_1(q_2)\Psi_1(q_3)$，$\Psi_2(q_1)\Psi_2(q_2)\Psi_2(q_3)$

兩個粒子位於同一態上：

$$\frac{1}{\sqrt{3}}\left(\Psi_i(q_1)\Psi_i(q_2)\Psi_j(q_3) + \Psi_i(q_1)\Psi_i(q_3)\Psi_j(q_2) + \Psi_i(q_3)\Psi_i(q_2)\Psi_j(q_1)\right),\quad i,j = 1,2,3$$

~~三個玻色子位於不同~~ ~~共 $3+A_3^2=9$ 種態~~ 共 $2+2! = 4$ 種態

7.(7.8) ~~[illegible]~~ 因為兩電子之間的庫侖勢很小，所以 $\hat{V} = U(\vec{r})$

對於單粒電子來說其波函數 $\Psi(x,y,z)$ 滿足

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + U(r)\right)\Psi = E\Psi, \quad U(r) = \frac{1}{2}m\omega^2 r^2 = \frac{1}{2}m\omega^2 x^2 + \frac{1}{2}m\omega^2 y^2 + \frac{1}{2}m\omega^2 z^2$$

$$= U(x) + U(y) + U(z)$$

$$\Rightarrow \hat{H} = -\frac{\hbar^2}{2m_e}\nabla^2 + U(r)$$

$$= \left[-\frac{\hbar^2}{2m_e}\frac{\partial^2}{\partial x^2} + U(x)\right] + \left[-\frac{\hbar^2}{2m_e}\frac{\partial^2}{\partial y^2} + U(y)\right] + \left[-\frac{\hbar^2}{2m_e}\frac{\partial^2}{\partial z^2} + U(z)\right]$$

$$= \hat{H}_x + \hat{H}_y + \hat{H}_z$$

利用分離變量法可得 $\Psi(x,y,z) = \varphi_{n_x}(x)\,\varphi_{n_y}(y)\,\varphi_{n_z}(z)$

其中 $\varphi_{n}(y)$ 滿足一維諧振子的 Schrödinger 方程

$$-\frac{\hbar^2}{2m_e}\frac{\partial^2}{\partial y^2}\varphi_i + \frac{1}{2}m\omega^2 y^2 = E_i\,(n+\tfrac{1}{2})\hbar\omega\,\varphi_i, \quad \varphi_i(y) = \sqrt{\frac{\alpha}{\pi^{\frac{1}{2}}\cdot 2^n n!}}\, e^{-\frac{1}{2}\alpha^2 y^2} H_n(\alpha y)$$

$$\alpha = \sqrt{\frac{m_e\omega}{\hbar}}$$

對於基態來說，$\Psi_{000}(x,y,z) = \left(\frac{\alpha}{\pi^{\frac{1}{2}}}\right)^{\frac{3}{2}} e^{-\frac{1}{2}\alpha^2 r^2}$

對於沿 x 方向運動的第一激發態來說，$\Psi_{100}(x,y,z) = 2\alpha x\cdot\left(\frac{\alpha}{\pi^{\frac{1}{2}}}\right)^{\frac{3}{2}} e^{-\frac{1}{2}\alpha^2 r^2}$

~~記兩個~~ 記兩個電子的位置和自旋態是 (x_1, y_1, z_1, m_{s1})，(x_2, y_2, z_2, m_{s2})

~~$\Psi_{xyz} = \frac{1}{\sqrt{2}}($~~

空間波函數可以組成一個對稱和反對稱函數.

$$\Psi_{S,r} = \frac{1}{\sqrt{2}}\left(\Psi_{000}(x_1,y_1,z_1)\Psi_{100}(x_2,y_2,z_2) + \Psi_{100}(x_1,y_1,z_1)\Psi_{000}(x_2,y_2,z_2)\right)$$

$$= \frac{1}{\sqrt{2}}\left(\frac{\alpha}{\pi^{\frac{1}{2}}}\right)^3\left[2\alpha x_2 e^{-\frac{1}{2}\alpha^2(r_1^2+r_2^2)} + 2\alpha x_1 e^{-\frac{1}{2}\alpha^2(r_1^2+r_2^2)}\right]$$

$$= \frac{1}{\sqrt{2}}\left(\frac{\alpha}{\pi^{\frac{1}{2}}}\right)^3\cdot 2\alpha(x_1+x_2)\exp\left[-\frac{1}{2}\alpha^2(r_1^2+r_2^2)\right]$$

$$\Psi_{A,r} = \frac{1}{\sqrt{2}}\left(\Psi_{000}(\vec{r}_1)\Psi_{100}(\vec{r}_2) - \Psi_{000}(\vec{r}_2)\Psi_{100}(\vec{r}_1)\right)$$

$$= \frac{1}{\sqrt{2}}\left(\frac{\alpha}{\pi^{\frac{1}{2}}}\right)^{3}\cdot 2\alpha(x_2 - x_1)$$

自旋波函數也包含三種對稱態和一種反對稱態

$$\chi_S^{(1)} = \chi_{\frac{1}{2}}(s_{1z})\,\chi_{\frac{1}{2}}(s_{2z})$$

$$\chi_S^{(2)} = \chi_{-\frac{1}{2}}(s_{1z})\,\chi_{-\frac{1}{2}}(s_{2z})$$

$$\chi_S^{(3)} = \frac{1}{\sqrt{2}}\left[\chi_{\frac{1}{2}}(s_{1z})\chi_{-\frac{1}{2}}(s_{2z}) + \chi_{-\frac{1}{2}}(s_{1z})\chi_{\frac{1}{2}}(s_{2z})\right]$$

$$\chi_S^{(4)} = \frac{1}{\sqrt{2}}\left[\chi_{\frac{1}{2}}(s_{1z})\chi_{-\frac{1}{2}}(s_{2z}) - \chi_{-\frac{1}{2}}(s_{1z})\chi_{\frac{1}{2}}(s_{2z})\right]$$

由於電子是費米子，所以 ~~[illegible]~~ 波函數是反對稱的.

$$\psi_1 = \psi_{S,r}\cdot\chi_S^{(4)}$$

$$\psi_2 = \psi_{A,r}\cdot\chi_S^{(1)}$$

$$\psi_3 = \psi_{A,r}\cdot\chi_S^{(2)}$$

$$\psi_4 = \psi_{A,r}\cdot\chi_S^{(3)}$$

8. (補充題 1)

(1) 兩個不同粒子，共有 $3^2=9$ 種態. 態的集合是 $\{\phi_i\phi_j \mid i,j=1,2,3\}$

(2) 兩個全同Bose子：共有 $\binom{3}{1}+\binom{3}{2}=6$ 種態，~~態的集合是~~ $\cancel{\{\phi_i\phi_j \mid i>j,\ i,j=1,2,3\}}$

~~(3)兩~~

~~態的集合是~~ $\cancel{\{\frac{1}{\sqrt{2}}(\phi_i\phi_j+\phi}$

態的集合是 $\left\{\frac{1}{\sqrt{2}}\left(\phi_i(\vec{r}_1)\phi_j(\vec{r}_2)+\phi_j(\vec{r}_1)\phi_i(\vec{r}_2)\right) \,\middle|\, i\ge j,\ i,j\in\{1,2,3\}\right\}$

(3) 兩個全同Fermi子，共有 $\binom{3}{2}=3$ 種態. 態的集合是 $\left\{\frac{1}{\sqrt{2}}\left(\phi_i(\vec{r}_1)\phi_j(\vec{r}_2)-\phi_i(\vec{r}_2)\phi_j(\vec{r}_1)\right) \,\middle|\, i>j,\ i,j\in\{1,2,3\}\right\}$

9. (補充題 2)

$$\overline{S_x}=(\chi_{-\frac{1}{2}},\ \hat{S}_x\chi_{-\frac{1}{2}})=(\chi_{-\frac{1}{2}},\ \tfrac{1}{i\hbar}[\hat{S}_y,\hat{S}_z]\chi_{-\frac{1}{2}})=\frac{1}{i\hbar}(\chi_{-\frac{1}{2}},\ (\hat{S}_y\hat{S}_z-\hat{S}_z\hat{S}_y)\chi_{-\frac{1}{2}})$$

$$=\frac{1}{i\hbar}\left[(\chi_{-\frac{1}{2}},\ -\tfrac{\hbar}{2}\hat{S}_y\chi_{-\frac{1}{2}})-(\hat{S}_z\chi_{-\frac{1}{2}},\ \hat{S}_y\chi_{-\frac{1}{2}})\right]$$

$$=\frac{1}{i\hbar}\left[-\frac{\hbar}{2}\overline{S_y}+\frac{\hbar}{2}\overline{S_y}\right]=0$$

$$\hat{S}_x^2=\frac{\hbar^2}{4}\ \Rightarrow\ \overline{S_x^2}=\frac{\hbar^2}{4}\ \Rightarrow\ \overline{\Delta S_x^2}=\frac{\hbar^2}{4}$$

同理可得 $\overline{S_y}=0$, $\overline{\Delta S_y^2}=\frac{\hbar^2}{4}$

10. $\hat{H}_{so}=A\hat{\vec{L}}\cdot\hat{\vec{S}}=\frac{1}{2}A\left[(\hat{\vec{L}}+\hat{\vec{S}})^2-\hat{\vec{L}}^2-\hat{\vec{S}}^2\right]=\frac{A}{2}\left[\hat{\vec{J}}^2-\hat{\vec{L}}^2-\hat{\vec{S}}^2\right]$

選定 $\{\hat{H}_{so},\ \hat{\vec{J}}^2,\ J_z,\ \hat{\vec{L}}^2,\ \hat{\vec{S}}^2\}$ 為力學量完全集，本征函數為 $|j m_j l s\rangle$

$$\hat{H}_{so}|j m_j l s\rangle=\frac{A}{2}\left[j(j+1)-l(l+1)-s(s+1)\right]\hbar|j m_j l s\rangle$$

$$E_{jls}=\frac{A}{2}\left[j(j+1)-l(l+1)-s(s+1)\right]$$，其中 $l=2$, $s=1$, $j=1,2,3$

$$E_j=\begin{cases}-3A\hbar^2, & j=1,\ \text{簡并度為 }3\\ -A\hbar^2, & j=2,\ \text{為 }5\\ 2A\hbar^2, & j=3,\ \text{為 }7\end{cases}$$